in-cosmetics® global
Innovation Zone Best Ingredient Award 2019
Finalist
Active Ingredient

Care Creations™ | BASF We create chemistry

Bix'Activ®

The cutaneous diet for a pure and mattified skin

Beauty Care Solutions Asia Pacific
Eunice Jeong / Emilly Shao

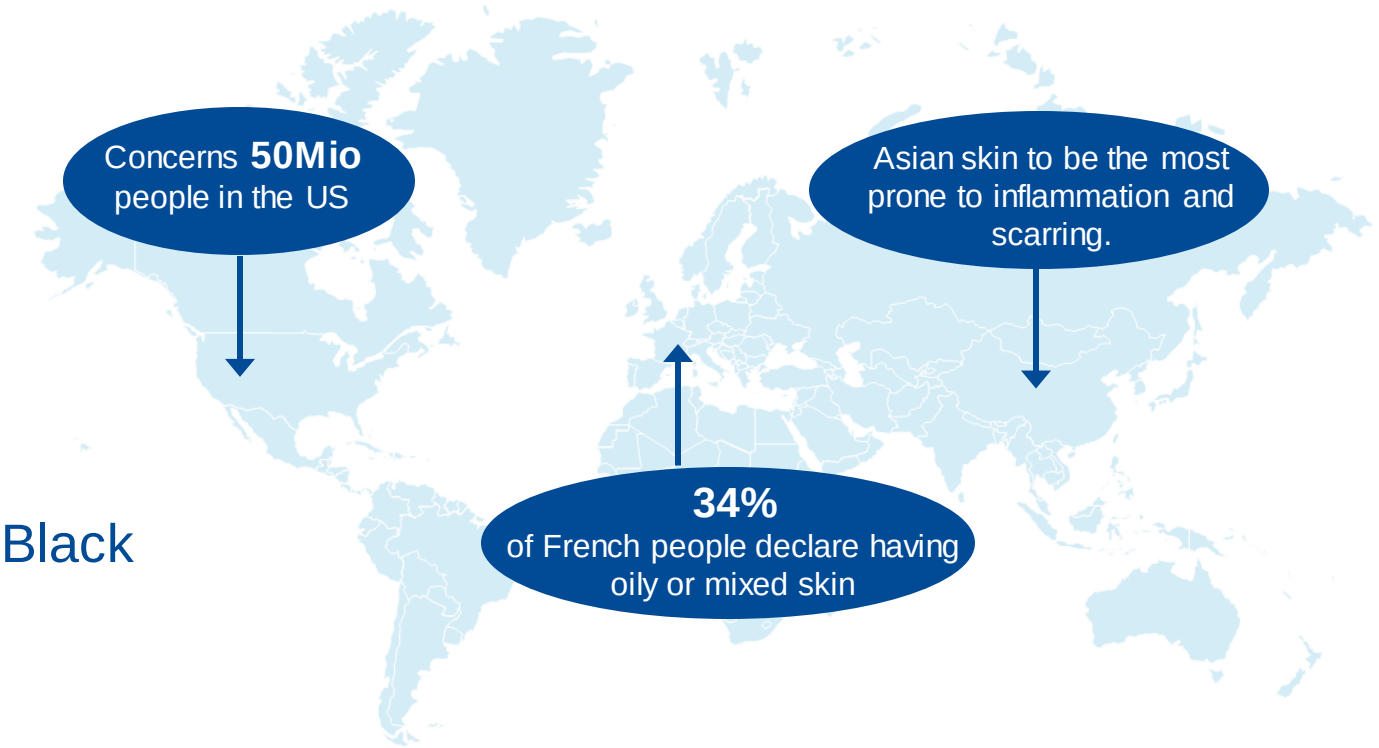
Korea | Sep 20, 2019



Oily skin: a global concern



- All skin types: Caucasian, Asian, Black
- All ages
- Both genders



Oily skin with imperfections is the biggest skin concern worldwide

▪ Internal



UNHAPPY WITH YOUR OILY COMPLEXION ?

Contrary to common belief, oily skin does not only concern teenagers

58%

of women worldwide between 25 and 34 are concerned by oily skin

MAIN CAUSES

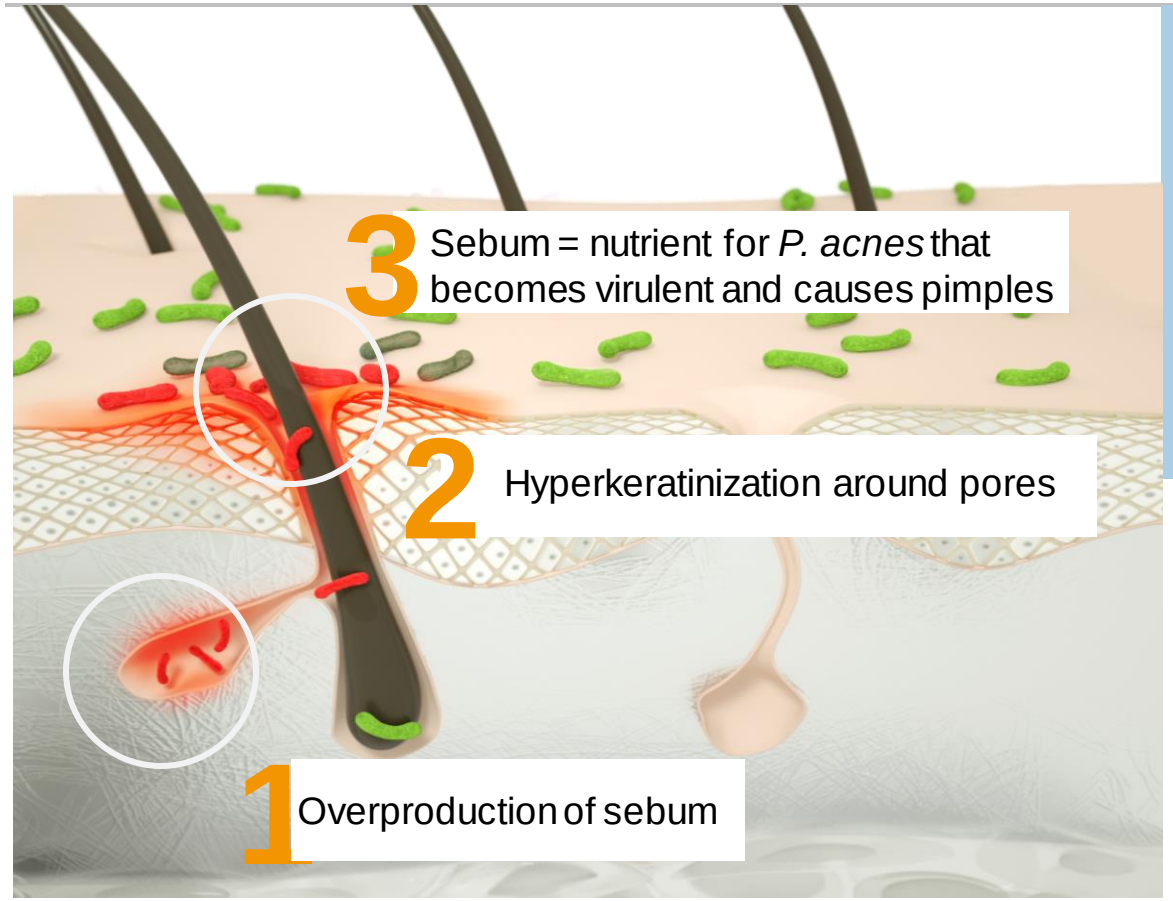
- Stress
- Treatments
- External factors
- Diet

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Source: Ipsos Beauty Track 2012

ernal

The origin of oily skin linked to sebaceous glands



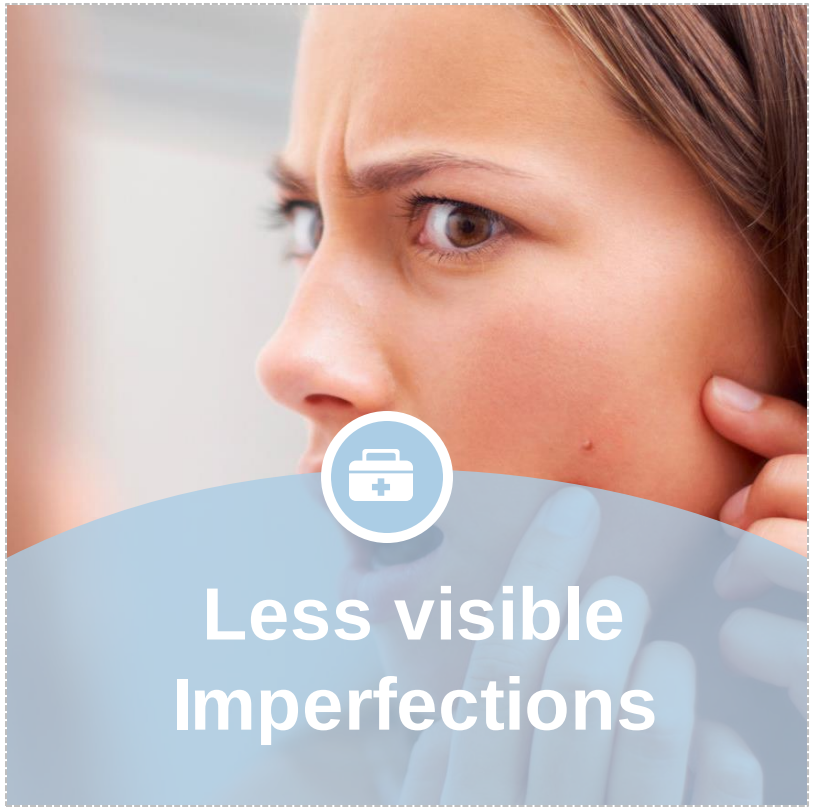
Oily skin is linked to hyperactive sebaceous glands that produce too much sebum

Over production of sebum	Hyperkeratinization	*P acnes* virulence causing inflammation
Shininess	Enlarged pores	Imperfections

Three clinical signs solved with Bix'Activ®





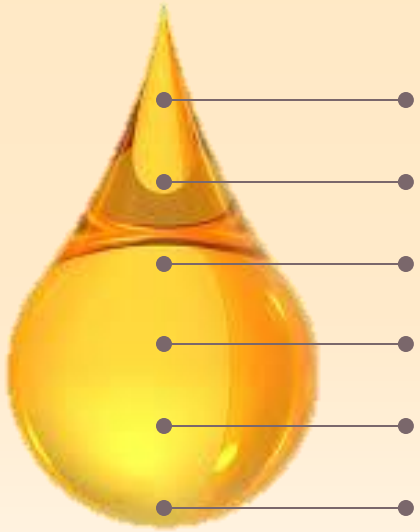


Skin hydration preserved

- Internal

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Sebum composition: complex mixture of lipids / Sebum functions

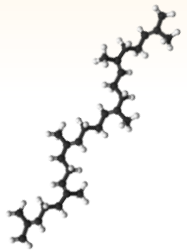


Triglycerides	30–50
Free Fatty Acids	**15–30**
Wax esters	**26–30**
Squalene	**12–20**
Cholesterol esters	3.0–6.0
Cholesterol	1.5–2.5

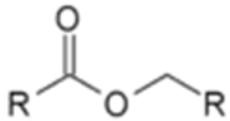
- Squalene and Wax esters not found anywhere else in the body nor among the epidermal surface lipids.
- Squalene is considered as a marker for sebogenesis

Functions attributed to sebum in humans:

- Photoprotection,
- Antimicrobial activity,
- Delivery of fat-soluble anti-oxidants to the skin surface,
- Pro- and anti-inflammatory activities
- If overproduction: Pathology



Squalene

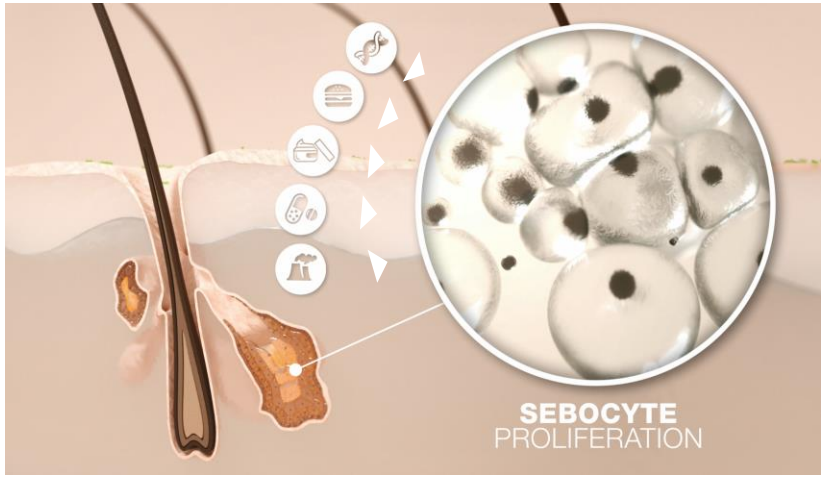


Wax ester

Internal

Shininess

Bix'Activ® helps reduce the activity of sebaceous gland for a mattifying effect – Multi-ethnic solution



Oily skin is linked to hyperactive sebaceous glands that produce excessive sebum

Sebum production *in vitro* with Bix'Activ® 0.02%

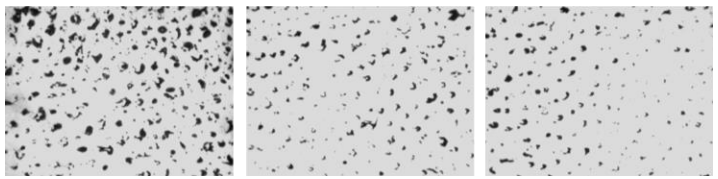
- African American sebocytes: -28%
- Caucasian sebocytes: -11%
- Asian sebocytes: -17%

Sebum production stimulated by HS Bix'Activ® at 0.02%



With Bix'Activ® at 0.25% 56 days vs D0 *in vivo*

- **Number of active pores reduced by :** -53%
- **Sebum excretion rate reduced by:** -71%



Measure of sebum production with Sebutape® method

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Internal

Visible action of Bix'Activ® on skin shininess

Shininess

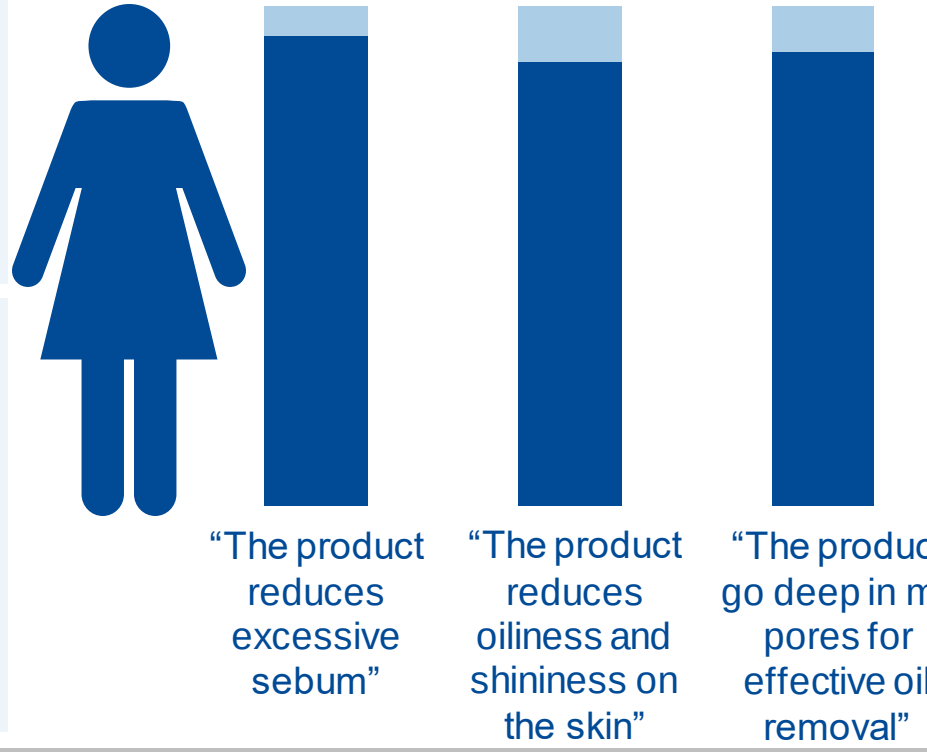
D0



D56



Self assessment



Internal

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Bix'Activ® acts like a cutaneous diet to block the signal inducing hyperkeratinization and avoid pores to appear bigger

Enlarged pores





- Our diet has an impact as well on keratinocytes due to IGF-1 production.
- IGF-1 stimulates the keratinocytes proliferation

Enlarged pores are favored by hyperproliferation of keratinocytes causing hyperkeratinization



Bix'Activ® 0.05% *in vitro*

Bix'Activ® stimulates IGFBP3 synthesis in keratinocytes to neutralize IGF-1

X1.7

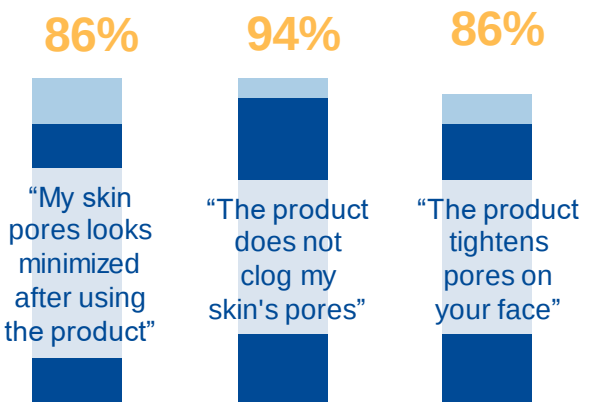
Bix'Activ® 0.25% *in vivo*

Pores 'size is decreased by

9%

Self assessment



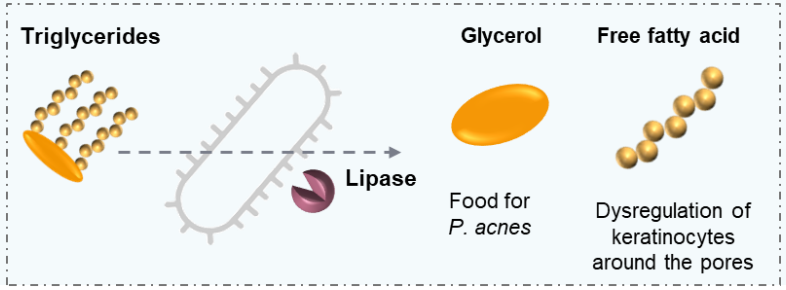


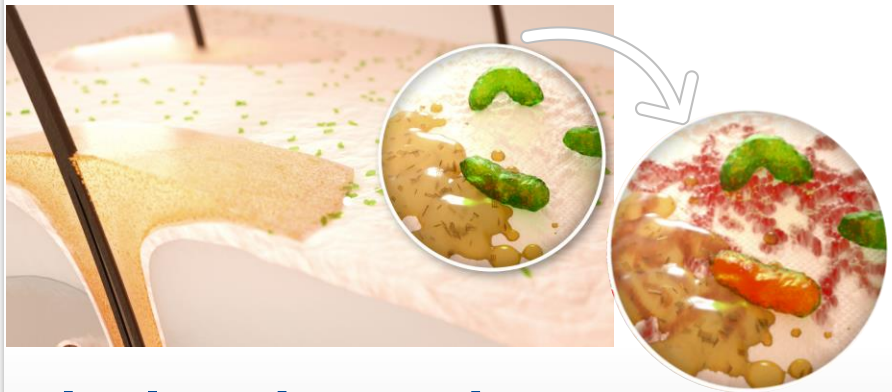
Internal

Imperfections

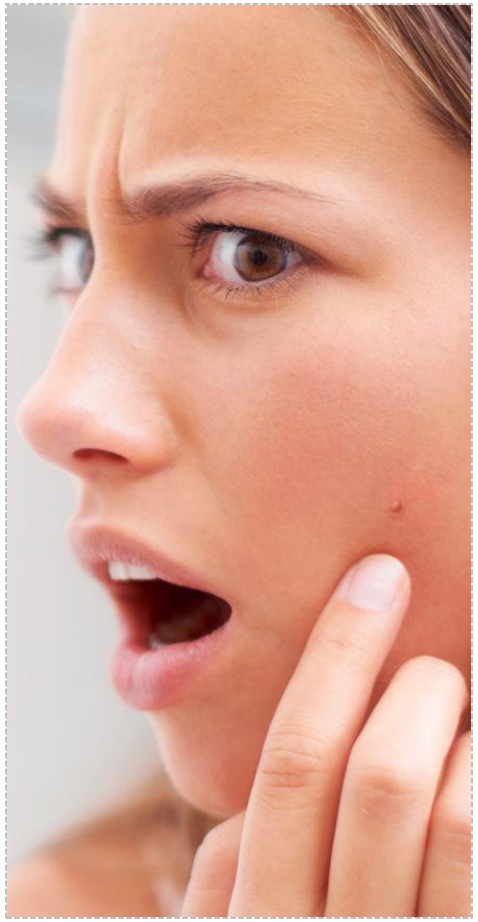
Bix'Activ® action on skin imperfections

Lipase released by *P. acnes* triggers the degradation of triglycerides





Virulent bacteria secrete lipase that triggers inflammation and create imperfections



In vitro

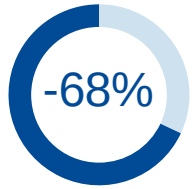
P. acnes with Bix'Activ® at 0.02%



4 days

37°C in anoxia

- *P. acnes virulence:*



Without direct effect on bacteria growth

In vivo

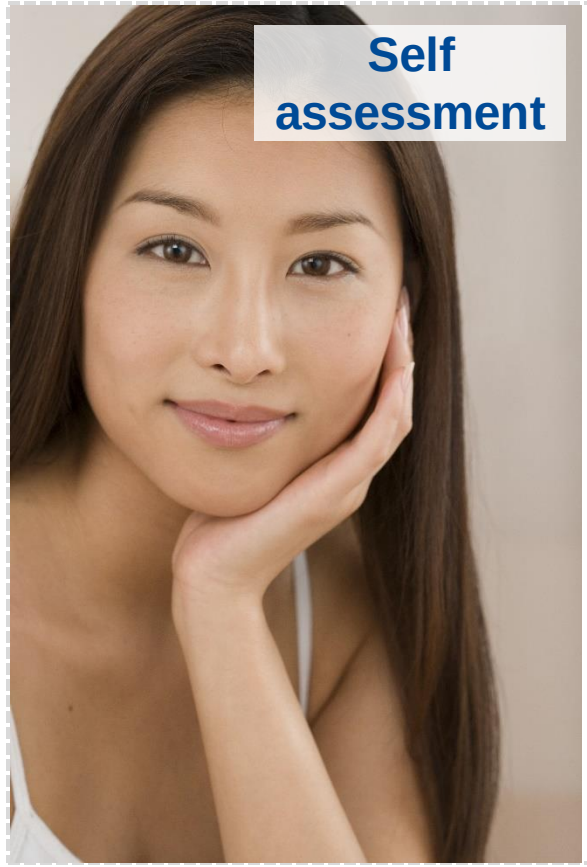
Hemi-face treated with Bix'Activ® at 0.25%

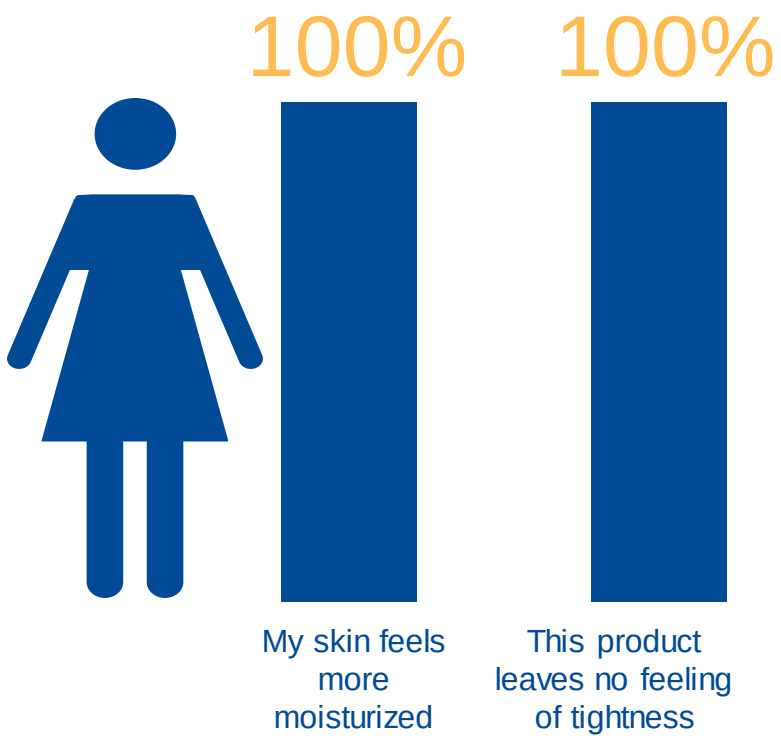


Visible reduction of skin imperfections appearance with results already after 4 weeks of treatment

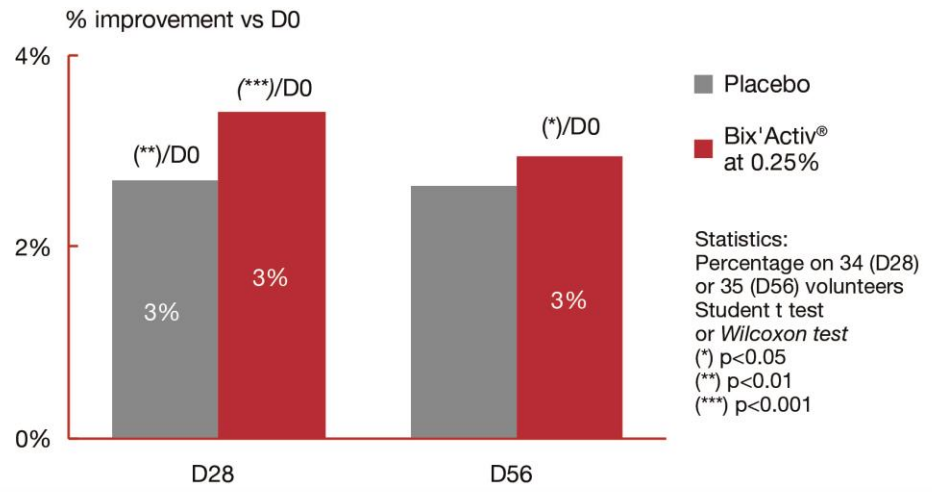
Internal

Bix'Activ® preserves skin moisturization





Hydration measurement with corneometer



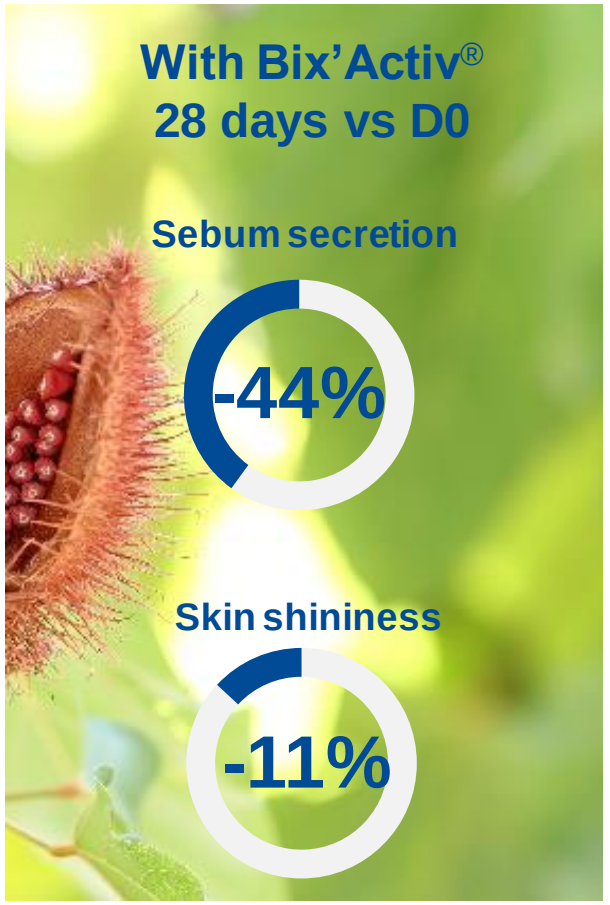
Bix'Activ® preserves the skin moisturized

Despite its strong efficacy on sebum regulation, Bix'Activ® doesn't dry out the skin that remains moisturized

▪ Internal

Bix'Activ® efficacy on back skin





93% of panelist confirmed their skin was less oily after 28 days of using Bix'Activ®

▪ Internal

Bixa orellana,

The red lip tree

INCI: Bixa orellana seed extract

Family: Bixaceae

Origin: Burkina Faso

Species: *Bixa orellana*

Plant part: Seed

Description: *Bixa orellana* is an evergreen shrub up to 8 m high, it bears clusters of white to pink flowers, comparable to wild roses. The non edible fruits are in clusters of red-brown seed pods covered in soft spines. Each pod contains many seeds covered with a thin waxy red aril rich in carotenoids. At maturity, the pod dries, hardens, and splits open, exposing the seeds.



Since ancient Mayan civilization, indigenous tribes of the South American rainforest have relied on *Bixa Orellana's* bright red annatto seeds for fabric dye, body paint and arts & crafts. In addition, tribes have long since believed that the seeds to work as an expectorant, and the roots to aid in digestion. For this reasons, native South Americans administer the whole plant as a natural medicinal treatment for bladder conditions, fevers, dysentery, burns, inflammation, heartburn and various skin problems.

Humid tropics

Native to tropical America but grows in other regions of South and Central America and Africa.

Sunny places

0 - 20 deg. C

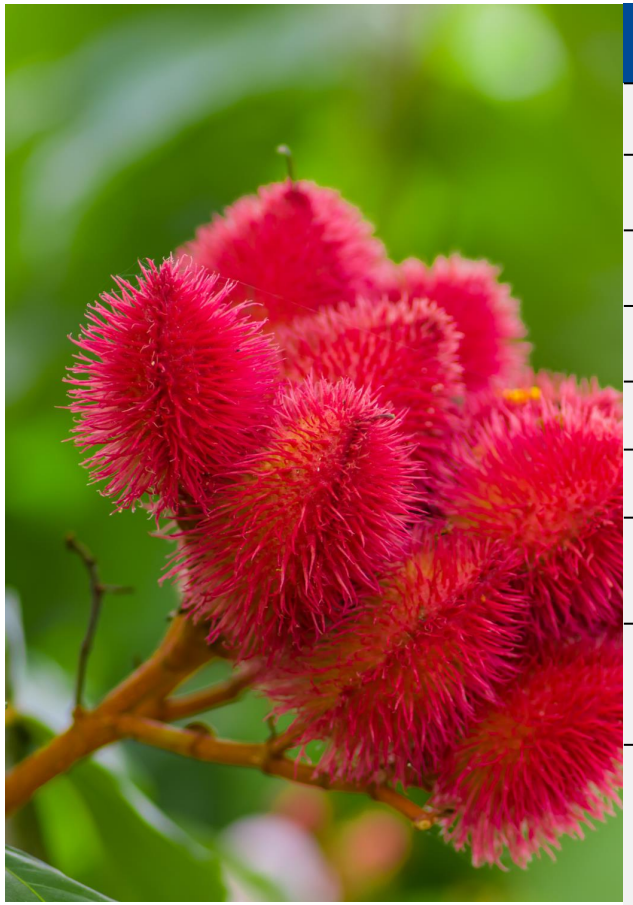
600- 2000 mm annual precipitation

The fruits are collected from November to December on domesticated trees in the west part of Burkina Faso. Then the fruits are manually opened to release the seeds. The seeds are sun dried.

BASF Partner sources the seeds from an historical supplier in Burkina Faso; it monitors the harvest with local independent farmers.

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Bix'Activ®
Product characteristics



Bix'Activ®	BC10050
Description	Standardized natural extract of *Bixa Orellana* seed, preservative free
INCI	Maltodextrin (and) Bixa Orellana Seed Extract
Composition 100%	50-70% Maltodextrin, 30-50% Bixa Orellana Seed Extract
Dose of use	0.25%
Aspect	Beige to yellowish fine powder
Solubility	Soluble in water
Preservative system	None
Naturalness content (ISO 16128)	100% natural origin
Regulatory data	PCPC ✓ / IECSC ✓ / IECIC 2015 ✓ / JP cosmetics ✓ / QD additive

China IECIC listed ✓

▪ Internal

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Lipase inhibition for oily skin

VISIBLE REDUCTION OF SEBUM EXCRETION AND PORE APPEARANCE

-62% Surface of spots

-37% Number of spots

-9% Pore size

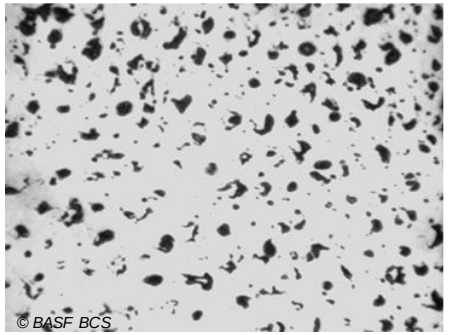
SKIN PERFECTION

-68% *P. acnes* virulence
Without direct effect on bacteria growth

Multi-ethnic efficacy

Sebum excretion rate

D0



D56

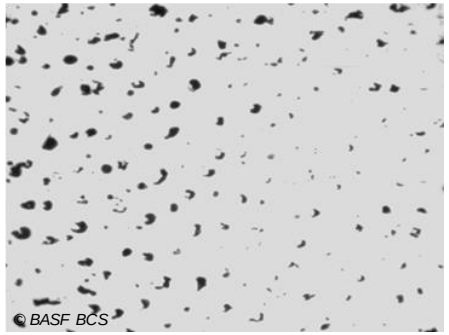
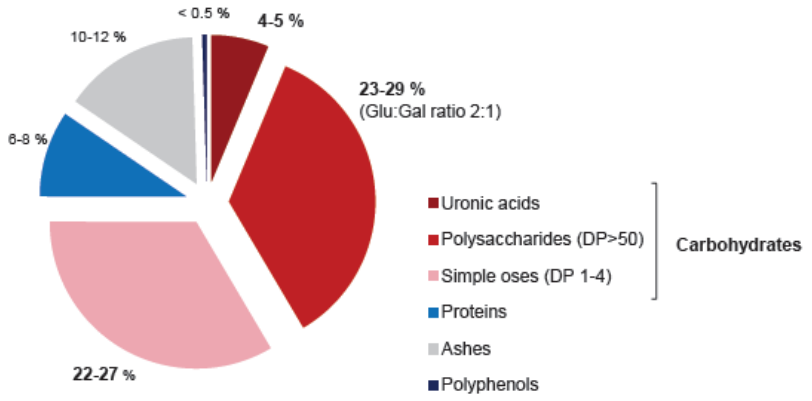


Image of decrease of surface and number of spots



Typical phytochemical composition of the plant extract



Bix'Activ ®

BASF

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